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DOCCONNECTS: REAL-TIME MEDICAL RISK ANALYZER

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1. ABSTRACT

DocConnects is a real-time medical risk analysis tool developed using core Machine Learning (ML) techniques. It provides early-stage risk prediction for heart attacks and diabetes by analyzing user inputs through an adaptive questionnaire system. Unlike traditional AI chatbots, this system primarily leverages statistical learning models, feature engineering, and real-world datasets to compute risk scores with high interpretability and precision.

Key ML components include:

- Predictive models (Logistic Regression, Decision Trees)
- Real-time feature evaluation
- Dynamic input handling via rule-based branching
- Scikit-learn-based classification engine

2. INTRODUCTION

Machine Learning has transformed predictive analytics in healthcare. DocConnects is developed as an ML-first application for proactive risk detection. The system's primary focus is on structured data ingestion, transformation, and classification to provide quantifiable risk scores.

The project is designed to give third-year students practical experience with supervised ML algorithms, feature importance analysis, and performance evaluation metrics such as accuracy, precision, and recall

3. EXISTING SYSTEMS

Many platforms like Ada Health and Buoy Health use general AI approaches, often focused on conversation and NLP. DocConnects differentiates itself by being data-driven and classification-focused.

Key limitations of existing systems:

- Lack of structured ML-based interpretability
- Focus on symptom collection rather than statistical prediction
- Static flows without dynamic risk re-evaluation

4. PROPOSED SYSTEM

4.1 ML-Driven Risk Engine

The heart of DocConnects is a real-time ML classification system. It works through the following pipeline:

- Data Collection (dynamic questionnaire)
- Feature Engineering (e.g., age, BMI, symptom presence)
- Classification using Logistic Regression or Decision Trees
- Risk Scoring and interpretation

4.2 Frontend Integration

- Developed with Flask and Jinja2
- Real-time input processed into tabular format
- Speech-to-text via Vosk for accessibility
- Risk classification output returned and displayed

4.3 Backend Architecture

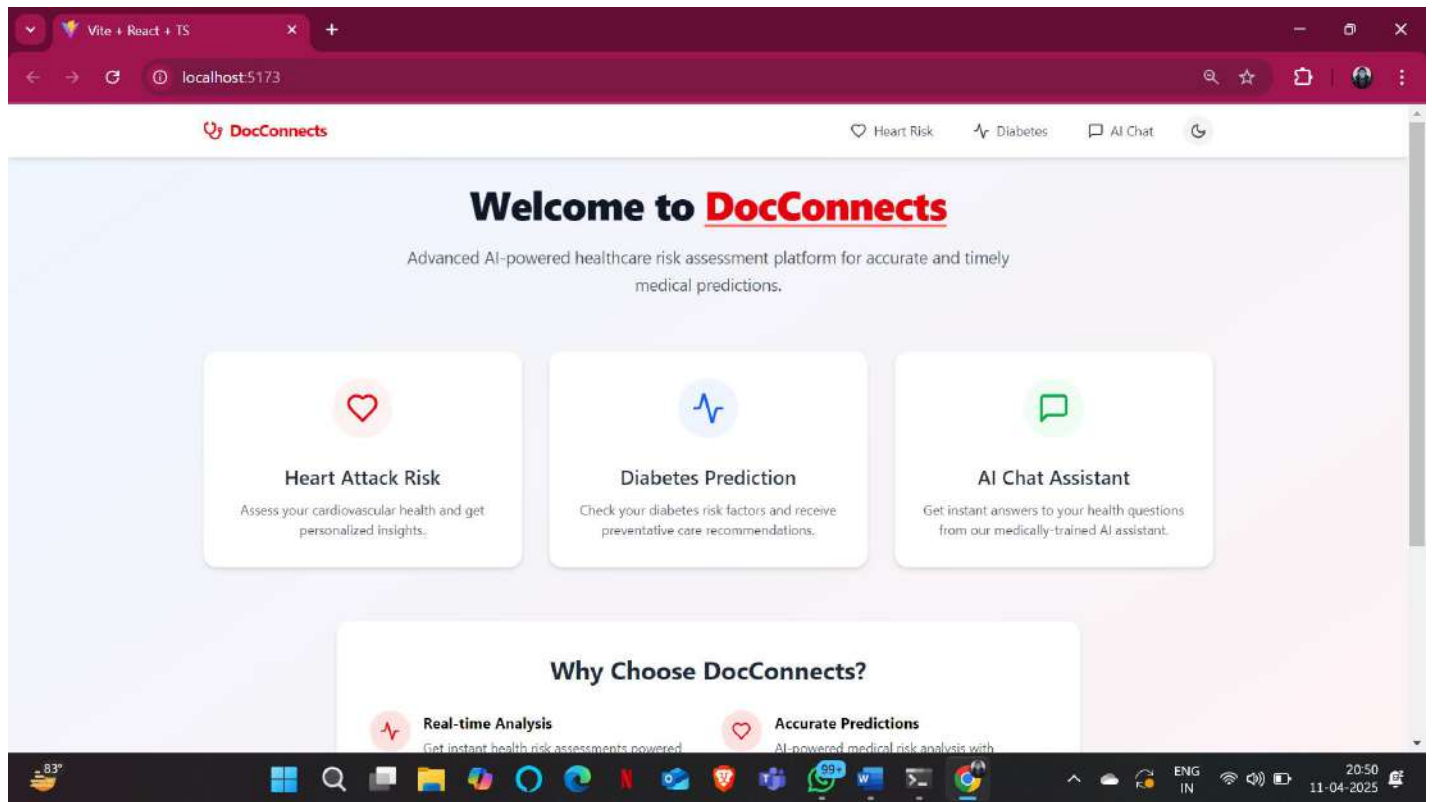
- Scikit-learn pipeline: preprocessing → feature scaling → classification
- DuckDuckGo integrated for dynamic question updates (not learning, just reference)
- REST-based endpoints for communication between front and back-end

5. DATA COLLECTION & MACHINE LEARNING APPROACH

Dataset Sources

- PIMA Indian Diabetes dataset
- Framingham Heart Study dataset
- Custom dataset from questionnaire responses
- Simulated datasets using Gaussian distributions for balance

6. SNAPSHOTS OF WORKING PROJECT





Heart Attack Risk Assessment

Fill out the form below with your health information to receive an assessment of your cardiovascular risk. All fields are required for an accurate prediction.

Personal Information

Age

21

Sex:

☒ Male

☐ Female

Clinical Measurements

Total Cholesterol (mg/dL)

220

LDL Cholesterol (mg/dL)

100

HDL Cholesterol (mg/dL)

130

Systolic Blood Pressure (mm Hg)

130

Diastolic Blood Pressure (mm Hg)

89.9

Lifestyle Factors

Smoking Status

☒ Yes

☐ No

Diabetes Status

☒ Yes

☐ No

Assess Heart Risk

Assessment Result

At risk

This assessment is based on the information you provided and should not replace professional medical advice. Please consult with a healthcare provider for a comprehensive evaluation.



Diabetes Risk Assessment

Fill out the form below with your health information to receive an assessment of your diabetes risk. All fields are required for an accurate prediction.

Personal Information

Sex:

☒ Male☐ Female

Age:

21

Clinical Measurements

Glucose Level (mg/dL)

180

Blood Pressure (mm Hg)

150

Skin Thickness (mm)

20

Insulin Level (μU/ml)

100

Additional Factors

BMI:

27

Diabetes Pedigree Function

1

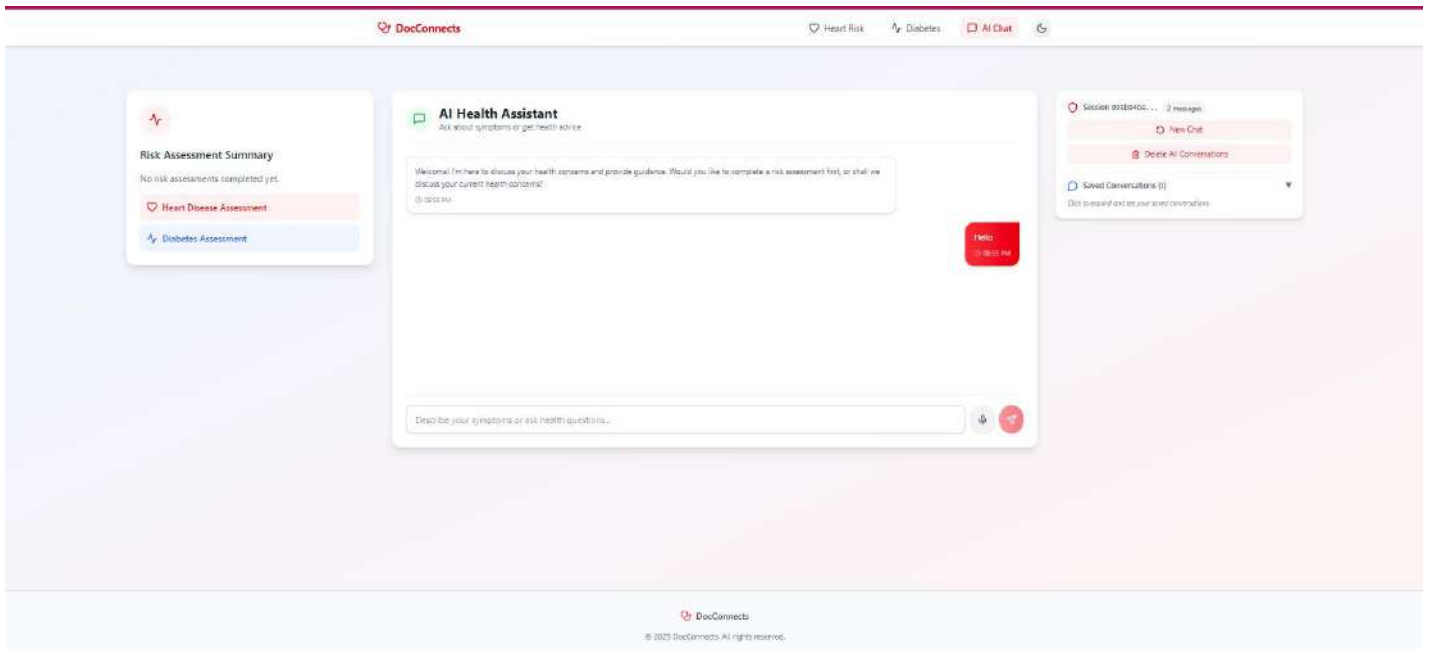
Indicates genetic influence for diabetes

Assess Diabetes Risk

Assessment Result

At risk

This assessment is based on the information you provided and should not replace professional medical advice. Please consult with a healthcare provider for a comprehensive evaluation.



7. FUTURE SCOPE

DocConnects emphasizes interpretable and statistically grounded machine learning for healthcare prediction. It uses lightweight yet powerful ML models that provide understandable and actionable results.

Future Directions:

- Expand ML models to multi-class risk detection (e.g., hypertension, thyroid)
- Incorporate advanced ensemble methods like XGBoost or CatBoost
- Create a feedback loop to collect real-world data for retraining
- Deploy as a scalable API for use by clinics and telemedicine apps

8. REFERENCES

- **Scikit-learn Documentation**
<https://scikit-learn.org/stable/documentation.html>
- **PIMA Indian Diabetes Dataset**
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- **Framingham Heart Study Dataset**
<https://www.kaggle.com/datasets/amanajmera1/framingham-heart-study-dataset>
- **Vosk Speech Recognition Toolkit**
<https://alphacephei.com/vosk/>
- **Research on Medical ML Interpretability**
<https://arxiv.org/abs/2010.15004>
- **Flask Documentation**
<https://flask.palletsprojects.com/>
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<https://www.hhs.gov/hipaa/index.html>
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